A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 - Generated practice paper Set 18

## Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance | Marks |
|---|---|---|
| **01.1** | **2 marks**<br>2 marks for the point below; award fewer where it is only partly made.<br>**2** When there is a difference in level between the sea and the basin: on the ebb (about 6-8 hours) water flows out through the turbines<br>– no generation when levels are equal (slack water)<br>**OR** possible on the flood when the sea is higher than the basin<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **01.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Two periods of slack water in each 12.5-hour cycle (twice in about 25 hours) give no flow<br>– the timing does not match demand<br>**1** Tides depend on the positions of the Moon and Sun which are known, so output can be forecast years ahead and alternative supplies planned<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **01.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1 Spring tides:** Sun and Moon aligned so their gravitational effects combine giving a large tidal range<br>– neap: effects oppose giving a small range<br>**1** Energy depends on the mass of water and height difference, so output is much greater at springs<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• The converse of any point, ie the same comparison written the other way round. | 2 |
| **01.4** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** When demand is low, surplus electricity pumps water into the lagoon above sea level<br>**1** The stored water is released through the turbines later when demand is high (pumped storage)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **pumped storage** → water pumped uphill when demand is low and released to generate at peak | 2 |
| **01.5** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Large tidal range (eg Severn Estuary)<br>**1** Estuaries, bays or headlands / channels that focus the flow and increase velocity<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 214-215; AQA spec 3.3.4.1. | |
| **02.1** | **Correct answer only**<br>1 mark for the correct option. No mark if more than one box is ticked.<br>**1** C<br>**Also accept**<br>• **stratospheric ozone** → the ozone in the stratosphere that absorbs UV | 1 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **02.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms<br>**1** Tropospheric ozone is a secondary pollutant (from NOx photochemistry) that is toxic to plants and humans and is a greenhouse gas<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• The converse of any point, ie the same comparison written the other way round.<br>• **Stratospheric ozone** → the ozone in the stratosphere that absorbs UV | 2 |
| **02.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere<br>**1** So most do not persist long enough to reach the stratosphere where UV would release their chlorine<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **CFCs** → CFCs / halogenated organic compounds releasing chlorine<br>**Do not credit**<br>• answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3. | |
| **03.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Readings drift so they may not be accurate<br>– calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions<br>**1** pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly<br>**1** If a meter cannot be adjusted, record the percentage error and correct the results<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **accurate** → close to the true value | 3 |
| **03.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Same height above the ground<br>**1** No obstacles to air flow<br>– impeller type facing into the wind<br>**1** Axis of rotation horizontal (impeller) or vertical (cup)<br>– same time period / averaging<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **03.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling<br>**1** Evaporation cools the wet bulb more when humidity is low<br>– the temperature difference is compared with a table to find humidity<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **03.4** | **Any 2 of 3**<br>1 mark each for any 2 of the 3 points below, in any order, maximum 2.<br>**1** Test strips with reagent bands compared with a colour chart (semi-quantitative)<br>**1 Electronic colourimeter:** reagent added and colour intensity measured at a specific wavelength<br>**1** Ion-selective electrode measuring electrode potential (other ions can interfere)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1. | |
| **04.1** | **Correct answer only**<br>1 mark for the correct option. No mark if more than one box is ticked.<br>**1** B - rivers are accessible over a large area with few natural contaminants | 1 |
| **04.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1 Potable / public supply:** no pathogens, low toxins, good appearance / taste / smell (public health)<br>**1 Industrial boiler / cooling water:** soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange<br>– spray irrigation: low turbidity to avoid blocked pipes and low heavy metals<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **04.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers)<br>**1 Disadvantage:** high cost / energy for pumping<br>– ecological impacts on the donor river<br>– transfer of pollutants or invasive species<br>– political conflict<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5. | |
| **05.1** | **Calculation**<br>A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark.<br>**1** (210 - 12) / 210 x 100<br>**1** = 94.3% (accept 94%)<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (%) is already printed on the answer line.<br>• **recycling** → reprocessing a material to make something new<br>**Do not credit**<br>• A bare wrong answer with no working.<br>• 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new | 2 |
| **05.2** | **Any 3 of 5**<br>1 mark each for any 3 of the 5 points below, in any order, maximum 3.<br>**1** Conserves mineral reserves / extends their lifetime<br>**1** Reduces the environmental impacts of mining (land take, habitat loss, spoil, dust, noise, turbid water)<br>**1** Reduces mineral processing pollution (eg red mud from bauxite refining)<br>**1** Reduces waste disposal impacts / landfill volume<br>**1** Less CO2 / climate change<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **recycling** → reprocessing a material to make something new<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.<br>• 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new<br>• 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **05.3** | **Any 4 of 5**<br>1 mark each for any 4 of the 5 points below, in any order, maximum 4.<br>**1 Identification of materials:** slow, labour-intensive to determine composition where it matters (eg alloys, plastic types)<br>**1 Separation of mixed materials:** alloys cannot easily be separated so recycled aluminium cannot be used where pure metal is needed<br>– mixed wastes reduce quality<br>**1** Reduction in quality / contamination (eg paper mixed with food waste)<br>**1 Transport / collection:** small dispersed quantities cannot use bulk transport so costs and energy per tonne are higher (may exceed savings)<br>**1** Lack of consumer cooperation in separating wastes<br>– waste losses to litter and general waste<br>– labour costs<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **recycling** → reprocessing a material to make something new<br>**Do not credit**<br>• Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.<br>• 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new | 4 |
| **05.4** | 1 mark for the point below.<br>**1** Designing products so that the materials can be re-used at the end of their lives - easy separation of components and identification of materials - extending the lifetime of mineral use<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 146-147; AQA spec 3.2.3.7. | |
| **06.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Shale has very low permeability so the 'tight' oil / gas cannot flow through the rock to a well<br>**1** Fluid is pumped down at high pressure to open fissures in the shale along which oil / gas can flow to the recovery well<br>**1** Sand grains (to hold fissures open) and solvents are pumped in to increase the recovery rate<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **hydraulic fracturing** → pumping fluid into rock at high pressure to release gas or oil<br>• **permeability** → how easily water passes through / lets water through | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **06.2** | **1 mark each**<br>1 mark for each of the 5 points below.<br>**1** Natural gas may enter aquifer water<br>**1** Injected chemicals may reach aquifers or the surface<br>– toxic metals in the rock may be mobilised<br>**1** Large volumes of water needed<br>**1** Earth tremors as fracking releases natural tensions in the crust (though not tremors that could not occur naturally)<br>**1 Reduced by:** collection, treatment and re-use of waste water<br>– well casing standards<br>– restricting sites away from aquifers / sensitive areas<br>– monitoring<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **fracking** → pumping fluid into rock at high pressure to release gas or oil<br>• **aquifer** → a permeable rock that stores groundwater / underground water store | 5 |
| **06.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Extraction is expensive and energy-intensive (quarrying and hot water treatment, steam / solvent / combustion in-situ; heating shale to release oil) so it is not economically viable at current prices<br>**1** Reserves only include what is economically exploitable now<br>– high environmental impacts (land take, water, CO2) also limit exploitation<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 184, 189-190; AQA spec 3.3.4.1. | |
| **07.1** | 1 mark for the point below.<br>**1** The time taken for half of a pollutant (released into the environment) to break down / degrade<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **half-life** → the time for the activity, or the amount, to fall by half<br>**Do not credit**<br>• 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues | 1 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **07.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** DDT is liposoluble so it is stored in fat / can pass through phospholipid cell membranes and is not excreted (nitrates are water soluble and excreted easily)<br>**1** DDT is persistent (long half-life) so it is not broken down before it is passed to the next trophic level<br>**1** So DDT bioaccumulates in each organism and becomes more concentrated at each successive trophic level<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **persistent** → does not break down / stays in the environment for a long time<br>• **half-life** → the time for the activity, or the amount, to fall by half<br>• **trophic level** → the feeding level in a food chain<br>**Do not credit**<br>• 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues<br>• 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts | 3 |
| **07.3** | **Any 2 of 3**<br>1 mark each for any 2 of the 3 points below, in any order, maximum 2.<br>**1** The gas is denser than air so it settled close to the ground / did not rise and disperse<br>**1** Low wind velocity (at night) meant it was not dispersed / diluted<br>**1** Short half-life so it broke down before travelling far<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **half-life** → the time for the activity, or the amount, to fall by half<br>**Do not credit**<br>• 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues | 2 |
| **07.4** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** High solubility means nitrates are easily dispersed / leached / carried by water so they affect a larger area (eg rivers, aquifers, the sea)<br>**1** But dispersal dilutes the pollutant so the concentration at any one point may be lower<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **leached** → nutrients or ions washed down out of the soil / washed away by water | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **07.5** | **Any 2 of 4**<br>1 mark each for any 2 of the 4 points below, in any order, maximum 2.<br>**1** Noise / sound<br>**1** Heat / thermal pollution<br>**1** Ionising radiation<br>**1** Light<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **Ionising radiation** → radiation with enough energy to remove electrons and form ions<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1. | |
| **08.1** | **1 mark each**<br>1 mark for each of the 5 points below.<br>**1 Terracing:** sloping land divided into narrow level fields held by retaining walls along contours<br>– water slows as it crosses each terrace and infiltrates instead of running off<br>**1 Tied ridging:** on nearly flat land a criss-cross of intersecting ridges retains rainwater, increasing infiltration and preventing runoff<br>**1 Windbreaks:** hedgerows / tree rows reduce wind velocity (for up to 30 times their height) so the wind has less kinetic energy to carry particles<br>– protect bare fields after harvest<br>**1 Mulching:** organic matter on the surface protects soil from raindrop impact and wind<br>– decomposes to humus which binds particles<br>– retains moisture<br>**1 Strip cropping:** strips of different crops harvested at different times (at 90° to the wind) so there is always a crop protecting part of the field<br>– slows runoff<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **Terracing** → cutting the slope into level steps<br>• **infiltration** → water soaking into the soil surface<br>• **humus** → dark, decomposed organic matter in soil<br>**Do not credit**<br>• 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself | 5 |
| **08.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** No frequent replanting / soil disturbance (ploughing) that breaks up soil structure and exposes particles<br>**1** Permanent root systems and canopy cover protect the soil all year<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **08.3** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1** Control stocking density so vegetation regrows faster than it is eaten<br>**1** Restrict access to vulnerable areas such as river banks (fencing)<br>**1** Move / remove livestock at high-risk times, eg after heavy rain when soil is wet<br>**1** Rotational grazing to rest pastures<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.<br>• 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself | 3 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 164-166; AQA spec 3.2.5.3. | |
| **09.1** | **Calculation**<br>1 mark for the correct answer; working does not have to be shown.<br>**1** 25 g $ha^{-1}$ (accept 24-27)<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (g $ha^{-1}$) is already printed on the answer line.<br>**Do not credit**<br>• A bare wrong answer with no working. | 1 |
| **09.2** | 1 mark for the point below.<br>**1** The highest dose that can be applied without killing any members of the population / any individuals of the (non-target) species<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| **09.3** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1** The MDAF (minimum dose always fatal) of the pest is about 50 g $ha^{-1}$<br>**1** This is higher than the MDNF of the bee (about 20 g $ha^{-1}$)<br>**1** So a dose that kills all pests kills some bees / the two curves overlap between about 20 and 50 g $ha^{-1}$<br>**1** A dose of 30-40 g $ha^{-1}$ would kill most pests with limited bee deaths (compromise)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **09.4** | **Any 2 of 5**<br>1 mark each for any 2 of the 5 points below, in any order, maximum 2.<br>**1** Spray at night / when bees are not flying<br>**1** Spray when the crop is not in flower<br>**1** Spray on still days to reduce spray drift<br>**1** Use a systemic pesticide / seed coating rather than spraying<br>**1** Use smaller droplets for more even coverage<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on | 2 |
| **09.5** | **Any 3 of 5**<br>1 mark each for any 3 of the 5 points below, in any order, maximum 3.<br>**1** Specificity is the extent to which the pesticide is toxic only to the target species / range of species affected<br>**1** A highly specific pesticide has low toxicity to non-target species (eg bees, fish, mammals)<br>**1** So fewer non-target deaths / less loss of predators, pollinators<br>**1** Fewer indirect (food chain) effects / fewer inter-species relationships lost<br>**1 Example:** pyrethroids have high insect toxicity but low mammal toxicity<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on<br>• 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts | 3 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-263; AQA spec 3.4.3.2.8. | |
| **10.1** | **Calculation**<br>1 mark for the correct answer; working does not have to be shown.<br>**1** 16.0 - 5.0 = £11 million<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (£ million) is already printed on the answer line.<br>**Do not credit**<br>• A bare wrong answer with no working. | 1 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **10.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** It is rarely practical to reduce emissions to zero and the impact of low-level emissions may be acceptable, so an acceptable level must be decided<br>**1 ALARA:** emissions should be As Low As Reasonably Achievable<br>**1 BATNEEC:** the Best Available Technology Not Entailing Excessive Cost is chosen to achieve this<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |
| **10.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** As industries expand or cities grow the number of pollutant sources increases<br>**1** So the total emissions / combined concentration rises even if each source is unchanged, and higher standards are needed<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **10.4** | **Levels of response**<br>Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist.<br>**Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.<br>**Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.<br>**Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.<br>**0 marks**: No relevant content.<br>**Indicative content - a guide to placing the mark, not a checklist**<br>• **Critical Pathway Analysis:** predicting movement of mobile persistent pollutants (state of effluent, wind, currents, geology, pH / oxygen and solubility, bioaccumulation, food sources, half-lives) so monitoring focuses on sites at risk<br>– mainly radioactive discharges<br>– checked by sampling<br>• **Critical Group Monitoring:** exposure of the most-exposed group monitored before health effects<br>– if their risk is acceptable everyone else is safer<br>– emissions reduced if not<br>• **Emission control strategies:** location (marine discharges, downwind of cities, not over aquifers) and timing (avoiding temperature inversions, low river flow)<br>• **Principles:** polluter pays (incentive depends on penalties and enforcement), precautionary principle (assume harm until shown otherwise; ignorance no excuse), ALARA / BATNEEC and the non-linear cost-efficiency relationship<br>• **Selection of methods:** production prevention (desulfurisation), prevention of release (precipitators, catalytic converters), post-release remediation (oil spill clean-up, phytoremediation), alternative processes (electric vehicles, renewables, pyrethroids)<br>• **Evaluation:** prevention usually cheaper and more effective than remediation<br>– costs rise steeply near 100%<br>– CPA relies on understanding of pathways<br>– enforcement and international agreement needed for transboundary pollutants<br>– judgement about acceptable risk is political as well as scientific<br>• Justified conclusion | 9 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | **Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• Any accurate, relevant material that is not in the indicative content.<br>• **persistent** → does not break down / stays in the environment for a long time<br>• **bioaccumulation** → a pollutant building up inside one organism over time<br>• **Critical Group** → the people most at risk / most heavily exposed<br>**Do not credit**<br>• answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain<br>• 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts | |
| | **Model answer - Level 3 (8 marks)** | |
| | Deciding how much to control pollution starts with predicting where a pollutant will go. Critical Pathway Analysis uses the state of the effluent, wind and currents, geology, the effect of pH and oxygen on solubility, bioaccumulation and local food sources to predict whether a discharge will disperse harmlessly or concentrate somewhere sensitive; it is mainly used for radioactive discharges and means monitoring can focus on the sites at risk rather than sampling everywhere. Critical Group Monitoring then checks the exposure of the people most at risk from their lifestyle before any health effects occur: if their risk is acceptable everyone else's is lower, and if not, emissions are reduced. | |
| | The choice of control is guided by principles. The polluter pays principle gives a financial incentive to prevent damage, but only if penalties are high and polluters cannot escape responsibility. The precautionary principle assumes a waste is harmful until research shows otherwise, so ignorance is no defence. Because the cost of control rises steeply as emissions approach zero, emissions are set As Low As Reasonably Achievable using the Best Available Technology Not Entailing Excessive Cost. Where emission location and timing can be chosen (marine outfalls, downwind of towns, not over aquifers; avoiding temperature inversions) severity is reduced cheaply. | |
| | Overall, prevention of production (fuel desulfurisation) or of release (electrostatic precipitators, catalytic converters) is more effective than post-release remediation such as oil-spill clean-up, which never recovers all the pollutant. CPA depends on understanding pathways, and transboundary pollutants need international enforcement, but together these principles give a rational basis for deciding acceptable risk rather than aiming for an unaffordable zero. | |
| | **Weaker answer - Level 1 (3 marks)** | |
| | Pollution can be controlled in lots of ways. Critical Pathway Analysis looks at where pollution goes and Critical Group Monitoring looks at people. The polluter pays principle means the person who pollutes has to pay. The precautionary principle means being careful. Pollution can be prevented, or cleaned up afterwards with things like booms for oil spills. Electric cars are an alternative process. Pollution control can be expensive so ALARA is used. All of these are good ways of controlling pollution. | |
| | **What makes the difference** | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | • The Level 3 answer explains *how* each principle works (why penalties matter, why costs rise near zero emissions); the weak answer only names them. | |
| | • The Level 3 answer groups the material logically (predict, decide, select) rather than listing everything at the same weight. | |
| | • 'Evaluate' needs comparison and a judgement: prevention vs remediation, CPA's dependence on understanding pathways, and a conclusion about acceptable risk. The weak answer ends with 'all of these are good'. | |
| | • Precise examples (desulfurisation, precipitators, catalytic converters, temperature inversions) show applied knowledge; 'things like booms' does not. | |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3. | |

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| Question | Marking guidance | Marks |
|---|---|---|
| 11.1 | **Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• **Nature of the problem:** interconnected atmosphere-hydrosphere-biosphere systems<br>– changes over different time and spatial scales<br>– natural fluctuations (solar output, orbit) masking or exaggerating anthropogenic change<br>– time delay between cause and effect (ocean heat capacity)<br>• **Limitations of historic data:** sparse global coverage, unsophisticated equipment, heat island bias, no atmospheric composition records<br>• **Proxy data and their reliability:** dendrochronology, coral growth rings, pollen in dated sediments, ice cores (gas bubbles, isotope ratios, volcanic ash layers, compaction problems)<br>• **Modern monitoring:** long-term data sets (Mauna Loa), electronic equipment, satellites (LEO vs geostationary), Argo floats, remote sensing of ice thickness / area / sea level<br>• Feedback mechanisms and tipping points adding uncertainty<br>– ecological uncertainty (species survival, distribution, fragmentation)<br>• **Computer models:** how they are tested and refined<br>– importance of accurate representative data<br>– scenario-based predictions and ranges | 25 |
| | • Uncertainty over ENSO / Conveyor / jet stream responses<br>• **Evaluation and conclusion:** confidence in the overall trend and human cause despite detailed uncertainty<br>– role of peer review and international bodies (IPCC)<br>*Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*<br>**Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90, 100-104, 10; AQA spec 3.2.1.2, 3.1.1.3. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| 11.2 | **Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• **Scale of the problem:** how energy is used (transport, buildings, industry, water)<br>– links between affluence and per capita use<br>• **Transport:** bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles<br>• **Buildings:** orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour<br>• **Industry:** heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management<br>• **Limits:** rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour<br>• Evaluation with quantitative examples<br>– comparison with the cost of new supply | 25 |
| | – conclusion that conservation is the cheapest first step but cannot alone meet demand growth<br>*Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*<br>**Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1. | |

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